

Retrofitting aging gas or oil boilers in small commercial properties with a sust

Comprehensive Hardware Design Analysis

Revision 1 | Generated 04/05/2026

Executive Summary

Generated

Project ID	a_compact_commercial_airsourse_heat_pump
Industry Domain	[LLM hypothesis] hvac_commercial
Engine Version	PDF Engine v2.1 (Full Detail)
Target Quantity	Not specified

ARCHITECTURE AT A GLANCE

MODULES [LLM hypothesis] 0	BILL OF MATERIALS ROWS [LLM hypothesis] 0	IDENTIFIED RISKS [LLM hypothesis] 0
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ECONOMICS

ESTIMATED UNIT COST [LLM hypothesis] £0.00	TARGET CEILING £4,500.00	HEADROOM [LLM hypothesis] £4,500.00
TOTAL NON-RECURRING ENGINEERING [LLM hypothesis] £0.00	UNIT COST PLUS AMORTISED NON-RECURRING ENGINEERING [LLM hypothesis] £0.00	

Cover

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Decision Required

BLOCKED — Brief Incomplete

Required field missing: product_type; Required field missing: target_cost; Required field missing: production_volume; Required field missing: max_mass; Required field missing: jurisdiction; Required field missing: thermal_capacity_kw; Required field missing: cop_target; Required field missing: refrigerant_type; Required field missing: acoustic_target_dba; Required field missing: architecture_type

Missing Required Inputs

FIELD	WHAT TO PROVIDE
product type	e.g. "air-source heat pump", "battery energy storage system"
target cost	Target unit manufacturing cost in GBP
production volume	Annual production quantity
max mass	Maximum allowable mass in kg
jurisdiction	Target market (e.g. "UK", "EU", "US")
thermal capacity kw	Heating/cooling capacity in kilowatts
cop target	Coefficient of Performance target at specified conditions
refrigerant type	e.g. "R290", "R410A", "R32"
acoustic target dba	Maximum noise level in dB(A)
architecture type	e.g. "monobloc", "refrigerant_split", "hydronic_split"

Next Actions

- Complete missing brief fields
- Lock architecture type
- Re-run pipeline

Biggest Blocker:

Required field missing: product_type

Contents

1. Brief & Requirements
2. Regulatory & Compliance Posture
3. Sizing & Spatial Optimisation
4. System Modules & Architecture
5. Bill of Materials
6. Cost Waterfall & Economics
7. Risks Register
8. Supplier Shortlist & Sourcing
9. Audit Log
10. Source Attribution

1. Brief & Requirements [Grade E — LLM hypothesis]

1.1 Overview & Context [Grade E — LLM hypothesis]

Subject / Use Case

Retrofitting aging gas or oil boilers in small commercial properties with a sustainable, high-efficiency hydronic heating system capable of a reliable 30kW thermal output.

Mission Statement

To accelerate the decarbonization of small UK commercial buildings through an affordable, ultra-low GWP, high-efficiency R290 heat pump solution.

Target Customers

Small commercial building owners, retail spaces, boutique hotels, and light industrial facilities in the UK.

Why Now (Market Timing)

Stringent UK F-gas regulations are rapidly phasing out high-GWP refrigerants, while fluctuating energy prices and national net-zero targets create intense demand for affordable, MCS-certified commercial heating alternatives.

Full Synthesis Report

The proposed product is a 30kW commercial air-source heat pump (ASHP) designed specifically for the UK small commercial market, striking a balance between high-capacity output, ultra-low global warming potential (GWP), and disruptive affordability. Utilizing R290 (propane) as its refrigerant, this unit targets an excellent Coefficient of Performance (COP) of ≥ 3.5 at standard A2/W35 conditions. This high efficiency relies on advanced inverter-driven compressors and optimized heat exchange surfaces, while completely eliminating the F-gas compliance burden for the end-user. The architecture is a hydronic split—keeping the highly flammable A3 R290 refrigerant safely contained within the outdoor unit to comply with EN 378 safety standards, while transferring thermal energy indoors via a safer water/glycol mixture to the indoor hydrobox. Achieving a strict unit target cost of £4,500 at an annual production volume of 500 units will be the primary engineering and supply chain challenge. It requires an aggressive reduction in Bill of Materials (BOM) costs, demanding standardized off-the-shelf components, optimized plate heat exchangers, and streamlined assembly

Data Sources:

[User] Founder brief text

[LLM] Gemini 3.1 Pro — research synthesis

Overall Source Grade: E

processes. Furthermore, securing stringent UK MCS (Microgeneration Certification Scheme) certification is critical, as it acts as a gatekeeper for government grants, commercial subsidies, and widespread market trust. Competitively, the UK market lacks a natively built, singular 30kW R290 unit at this disruptive price point, as most current installations rely on expensive, cascaded domestic units or over-specified industrial plants. This product effectively bridges the gap, empowering retail units, boutique hotels, and light industrial facilities to achieve net-zero heating economically, efficiently, and safely.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis
Overall Source Grade: E

1.2 Market & Competitors [Grade E — LLM hypothesis]

Competitor Landscape

Vaillant

Product	aroTHERM plus (Cascaded)
Pricing Strategy	£8,500+ (for two cascaded units)
Technical Specs	R290, 15kW per unit (needs 2 cascaded for 30kW), COP ~3.6 at A2/W35.

STRENGTHS

Exceptional brand trust, highly reliable, robust MCS track record.

WEAKNESSES

Requires cascading multiple residential units for commercial loads, high capital cost, larger combined footprint.

DIFFERENTIATION ANGLE

A native 30kW single-unit design significantly reduces installation complexity, footprint, and BOM cost compared to cascading smaller units.

Mitsubishi Electric

Product	Ecodan CAHV Series
Pricing Strategy	£12,000+
Technical Specs	40kW+ capacity, historically R407C/R454B moving to low-GWP, high efficiency.

STRENGTHS

Industrial-grade reliability, excellent BMS (Building Management System) integration, deep commercial installer network.

WEAKNESSES

Extremely high capital expense, historically reliant on higher GWP refrigerants, oversized for small boutique retail.

DIFFERENTIATION ANGLE

Targeting a disruptive £4,500 price point makes decarbonization accessible for small businesses unable to afford premium industrial tier systems.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Overall Source Grade: E

Clivet

Product	Thunder (R290 Commercial Heat Pump)
Pricing Strategy	£9,500
Technical Specs	Air-source, R290, 30kW to 85kW range capabilities, COP > 3.4.

STRENGTHS

High temperature output capabilities, scalable capacities na-
tively designed for commercial use.

WEAKNESSES

Lower brand awareness in the UK light-commercial sector,
relatively large footprint, expensive.

DIFFERENTIATION ANGLE

Hyper-focused on compactness and a highly optimized supply chain strictly geared towards the 30kW tier to achieve
a massive price advantage.

Data Sources:

[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Overall Source Grade: E

1.3 Engineering Constraints [Grade E — LLM hypothesis]

Core Targets

Unit Cost Ceiling	£4,500.00
Max Mass	-
Target Process	
Target Material	
Tolerance Target	
Production Quantity	
Batch Size	500

Research Sources

TITLE	TYPE	YEAR	RELEVANCE
UK MCS Standard MIS 3005-I	Regulatory Standard	2026	Outlines the installation and product requirements essential for MCS certification in the UK heat pump market.
BSRIA UK Heat Pump Market Report	Market Research	2026	Provides volumetric data, pricing benchmarks, and trend analysis for commercial ASHPs in the UK.
EN 378 Refrigerant Safety Standard	Safety Standard	2026	Mandates strict safety limits, charge sizes, and design constraints for using A3 highly flammable refrigerants like R290.
CIBSE TM51: Heating Design in Buildings	Design Guide	2026	Crucial for understanding thermal demands, flow temperatures, and sizing requirements for UK commercial properties.
Danfoss / Copeland Commercial Compressor Data	Component Specifications	2026	Determines the technical feasibility, cost, and efficiency of inverter-driven R290 compressors capable of 30kW output.

Data Sources:
[User] Founder brief text
[LLM] Gemini 3.1 Pro — research synthesis

Overall Source Grade: E

Note: Some regulatory assessments (charge-limit calculations, DSEAR classification) require completed sizing data. These will be populated once sizing is complete.

2.1 EN 378 [Grade E — LLM hypothesis]

Refrigerating systems and heat pumps - Safety

Status: not-started

Owner: Mechanical Engineering Lead

Standard Summary

Governs the safety and environmental requirements for refrigerating systems, specifically charge limits for A3 refrigerants.

Applicability Assessment

Outdoor unit refrigerant circuit and proximity to buildings.

Engineering Design Impact

Requires sealed outdoor containment, explosion-proof fan motors, and preventing R290 from entering the indoor hydrobox.

EVIDENCE REQUIRED

Leakage test reports, ATEX component certifications, and structural isolation proofs.

GAP ACTION

Design the hydronic split barrier and source ATEX-rated electrical components for the outdoor unit.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

Overall Source Grade: E

2.2 MCS 007 [Grade E — LLM hypothesis]

Heat Pump Product Certification Scheme

Status: not-started

Owner: Compliance Manager

Standard Summary

Establishing the performance, reliability, and manufacturing quality requirements to sell certified heat pumps in the UK.

Applicability Assessment

Entire system performance and factory production control.

Engineering Design Impact

Requires rigorous performance testing, COP validation, and factory ISO 9001 compliance for subsidized sales.

EVIDENCE REQUIRED

Approved laboratory test results, factory audit reports, and quality manuals.

GAP ACTION

Book initial pre-compliance testing with an MCS-accredited laboratory.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

Overall Source Grade: E

2.3 BS EN 14511 [Grade E — LLM hypothesis]

Air conditioners, liquid chilling packages and heat pumps

Status: not-started

Owner: Thermodynamics Engineer

Standard Summary

Specifies the test conditions for the rating of air-source heat pumps.

Applicability Assessment

Thermal performance specifications (COP >= 3.5 at A2/W35).

Engineering Design Impact

Dictates evaporator and condenser sizing, as well as compressor mapping to ensure nominal ratings are legally accurate.

EVIDENCE REQUIRED

Calibrated psychrometric chamber test data at specified temperature brackets.

GAP ACTION

Develop a thermodynamic model to simulate EN 14511 test conditions prior to prototyping.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

Overall Source Grade: E

2.4 ErP Directive (Lot 1) [Grade E — LLM hypothesis]

Ecodesign Directive for Space Heaters

Status: not-started

Owner: Systems Engineer

Standard Summary

Sets minimum seasonal space heating energy efficiency (SCOP) and sound power level limits.

Applicability Assessment

System control logic, physical dimensions, and acoustic design.

Engineering Design Impact

Requires advanced inverter control for high seasonal efficiency and specialized acoustic damping for the compressor/fan.

EVIDENCE REQUIRED

Acoustic test reports and SCOP calculation declarations.

GAP ACTION

Calculate projected SCOP and conduct early acoustic modeling of the chassis.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

Overall Source Grade: E

2.5 UKCA / CE Marking [Grade E — LLM hypothesis]

Electromagnetic Compatibility (EMC) and Low Voltage Directive (LVD)

Status: not-started

Owner: Electrical Engineering Lead

Standard Summary

Ensures electrical safety and prevents electromagnetic interference with other building systems.

Applicability Assessment

Inverter drives, control boards, and power supply units.

Engineering Design Impact

Mandates proper grounding, shielded cabling, and validated PCB designs.

EVIDENCE REQUIRED

EMC emissions/immunity test reports and LVD safety certificates.

GAP ACTION

Review inverter and controller supplier declarations of conformity and plan full system EMC testing.

Data Sources:

[LLM] Gemini 3.1 Pro — standards extraction

Overall Source Grade: E

3. Sizing

BLOCKED — Sizing not generated

Sizing requires: thermal capacity, ambient temperature, flow temperature, and envelope dimensions. These fields are missing from the brief.

4. System Modules

BLOCKED — Modules not generated

Module architecture requires completed sizing and spatial allocation.

5. Bill of Materials

BLOCKED — BOM not generated

Bill of materials requires completed module architecture.

6. Cost Waterfall

BLOCKED — Cost analysis requires completed sizing and bill of materials

Unit cost cannot be calculated without component selection. Complete sizing and BOM generation first.

7. Risks Register

BLOCKED — Risks not generated

Risk register requires completed system modules.

8. Supplier Shortlist

BLOCKED — Suppliers not matched

Supplier matching requires a completed bill of materials.

9. Audit Log

Pipeline execution log — generated by PDF Engine v2.1.

This document was generated automatically. Contents represent the synthesised output of the specialist pipeline.

Pipeline Execution Summary

Project Identifier	a_compact_commercial_airsource_heat_pump
Research Result	Generated
System Modules	0 Modules
Bill of Materials Lines	0 Parts
Cost Analysis	Pending
Specialist Reviews	0 Completed
Supplier Shortlists	0 Matched

10. Source Attribution

This table documents the origin of data for each major section of the report, ensuring traceability of LLM outputs versus deterministic or user-provided data.

SECTION	SOURCE	PROVIDER / SYSTEM	DETAIL
Research	LLM	OpenRouter	Gemini 3.1 Pro via OpenRouter
Research	USER	user	Founder brief text
Regulatory	LLM	Unknown LLM	Gemini 3.1 Pro — standards extraction